

# git log/history

## References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

## Git History - Directory Commit Statistics

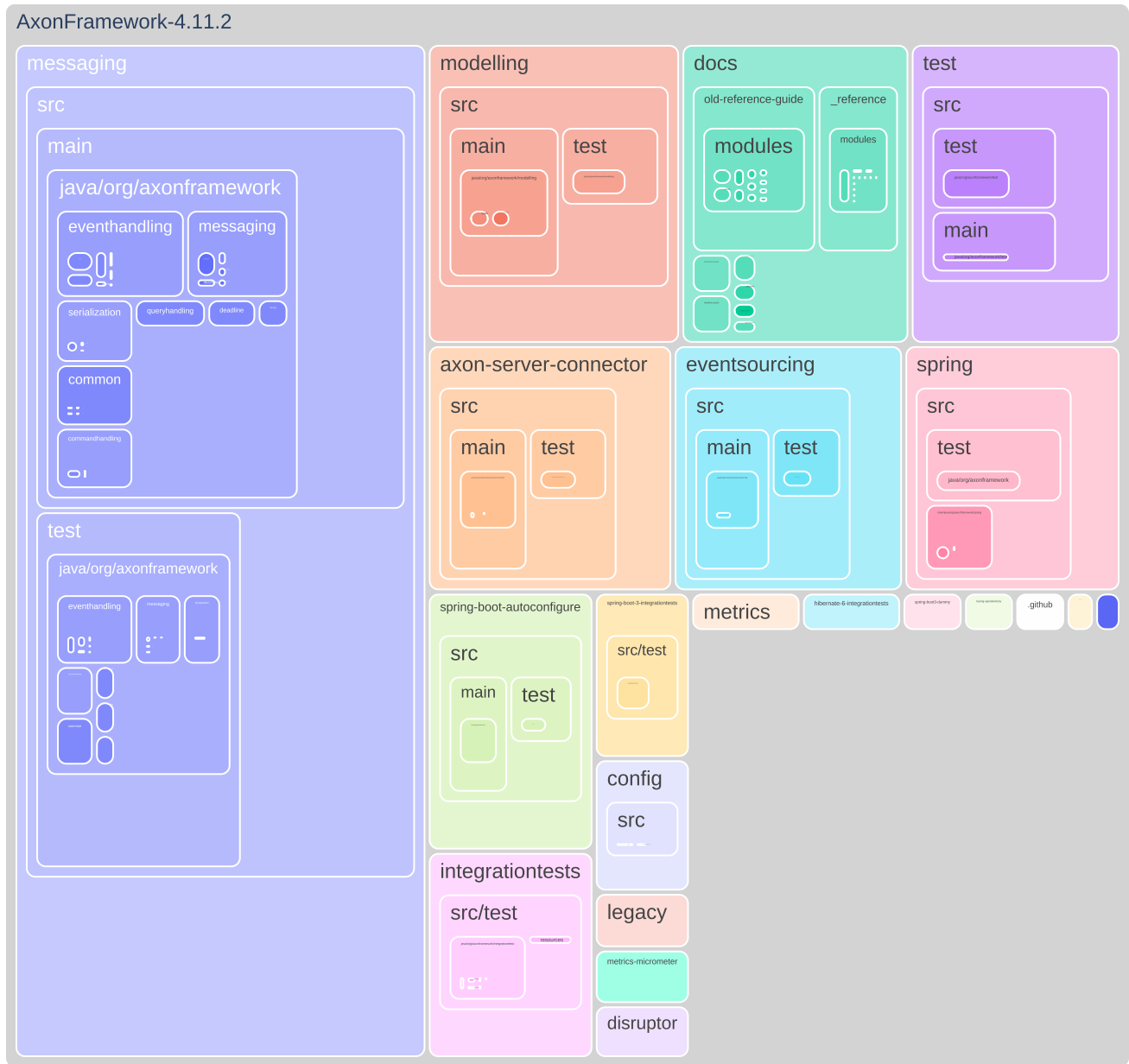
### Data Preview

|       | fileCount   | authorCount | commitCount | daysSinceLastCommit | daysSinceLastCreation | daysSinceLastModification |
|-------|-------------|-------------|-------------|---------------------|-----------------------|---------------------------|
| count | 563.000000  | 563.000000  | 563.000000  | 563.000000          | 563.000000            | 563.000000                |
| mean  | 24.492007   | 19.548845   | 283.804618  | 224.602131          | 1091.884547           | 317.534636                |
| std   | 117.928146  | 21.254240   | 630.075836  | 137.280170          | 968.702108            | 373.632167                |
| min   | 1.000000    | 2.000000    | 3.000000    | 25.000000           | 93.000000             | 24.000000                 |
| 25%   | 1.000000    | 7.000000    | 27.000000   | 151.000000          | 333.000000            | 150.000000                |
| 50%   | 4.000000    | 12.000000   | 80.000000   | 293.000000          | 852.000000            | 292.000000                |
| 75%   | 10.000000   | 25.000000   | 262.000000  | 293.000000          | 1906.000000           | 292.000000                |
| max   | 2250.000000 | 212.000000  | 9638.000000 | 2443.000000         | 5570.000000           | 2921.000000               |

|    | directoryPath                                                                                                    | directoryParentPat                                               |
|----|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 0  | AxonFramework-4.11.2/assembly                                                                                    | AxonFramework-4.11.                                              |
| 1  | AxonFramework-4.11.2/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source | AxonFramework-4.11.2/axon-serve                                  |
| 2  | AxonFramework-4.11.2/axon-server-connector/src/main/resources/META-INF/services                                  | AxonFramework-4.11.2/axon-server-connector/src/main/resource     |
| 3  | AxonFramework-4.11.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util       | AxonFramework-4.11.2/axon-serve                                  |
| 4  | AxonFramework-4.11.2/config/src/test/resources/META-INF                                                          | AxonFramework-4.11.2/config/src/test/resource                    |
| 5  | AxonFramework-4.11.2/coverage-report                                                                             | AxonFramework-4.11.                                              |
| 6  | AxonFramework-4.11.2/docs/_playbook/localLinks                                                                   | AxonFramework-4.11.2/docs/_playboc                               |
| 7  | AxonFramework-4.11.2/docs/_reference/modules/ROOT/attachments                                                    | AxonFramework-4.11.2/docs/_reference/modules/ROO                 |
| 8  | AxonFramework-4.11.2/docs/_reference/modules/ROOT/partials                                                       | AxonFramework-4.11.2/docs/_reference/modules/ROO                 |
| 9  | AxonFramework-4.11.2/docs/_reference/modules/axon-server-connector/pages                                         | AxonFramework-4.11.2/docs/_reference/modules/axon-server-connect |
| 10 | AxonFramework-4.11.2/docs/_reference/modules/axon-server-connector/partials                                      | AxonFramework-4.11.2/docs/_reference/modules/axon-server-connect |
| 11 | AxonFramework-4.11.2/docs/_reference/modules/configuration/pages                                                 | AxonFramework-4.11.2/docs/_reference/modules/configuratio        |
| 12 | AxonFramework-4.11.2/docs/_reference/modules/configuration/partials                                              | AxonFramework-4.11.2/docs/_reference/modules/configuratio        |
| 13 | AxonFramework-4.11.2/docs/_reference/modules/disruptor/pages                                                     | AxonFramework-4.11.2/docs/_reference/modules/disruptor           |
| 14 | AxonFramework-4.11.2/docs/_reference/modules/disruptor/partials                                                  | AxonFramework-4.11.2/docs/_reference/modules/disruptor           |
| 15 | AxonFramework-4.11.2/docs/_reference/modules/eventsourcing/pages                                                 | AxonFramework-4.11.2/docs/_reference/modules/eventssourcin       |
| 16 | AxonFramework-4.11.2/docs/_reference/modules/eventsourcing/partials                                              | AxonFramework-4.11.2/docs/_reference/modules/eventssourcin       |
| 17 | AxonFramework-4.11.2/docs/_reference/modules/legacy/pages                                                        | AxonFramework-4.11.2/docs/_reference/modules/legac               |
| 18 | AxonFramework-4.11.2/docs/_reference/modules/legacy/partials                                                     | AxonFramework-4.11.2/docs/_reference/modules/legac               |
| 19 | AxonFramework-4.11.2/docs/_reference/modules/metrics/pages                                                       | AxonFramework-4.11.2/docs/_reference/modules/metric              |
| 20 | AxonFramework-4.11.2/docs/_reference/modules/metrics/partials                                                    | AxonFramework-4.11.2/docs/_reference/modules/metric              |
| 21 | AxonFramework-4.11.2/docs/_reference/modules/micrometer/pages                                                    | AxonFramework-4.11.2/docs/_reference/modules/micromete           |
| 22 | AxonFramework-4.11.2/docs/_reference/modules/micrometer/partials                                                 | AxonFramework-4.11.2/docs/_reference/modules/micromete           |
| 23 | AxonFramework-4.11.2/docs/_reference/modules/modeling/partials                                                   | AxonFramework-4.11.2/docs/_reference/modules/modelin             |
| 24 | AxonFramework-4.11.2/docs/_reference/modules/spring/pages                                                        | AxonFramework-4.11.2/docs/_reference/modules/sprin               |
| 25 | AxonFramework-4.11.2/docs/_reference/modules/spring/partials                                                     | AxonFramework-4.11.2/docs/_reference/modules/sprin               |
| 26 | AxonFramework-4.11.2/docs/_reference/modules/springboot-starter/pages                                            | AxonFramework-4.11.2/docs/_reference/modules/springboot-starte   |
| 27 | AxonFramework-4.11.2/docs/_reference/modules/springboot-starter/partials                                         | AxonFramework-4.11.2/docs/_reference/modules/springboot-starte   |
| 28 | AxonFramework-4.11.2/docs/_reference/modules/test/pages                                                          | AxonFramework-4.11.2/docs/_reference/modules/te:                 |
| 29 | AxonFramework-4.11.2/docs/_reference/modules/test/partials                                                       | AxonFramework-4.11.2/docs/_reference/modules/te:                 |

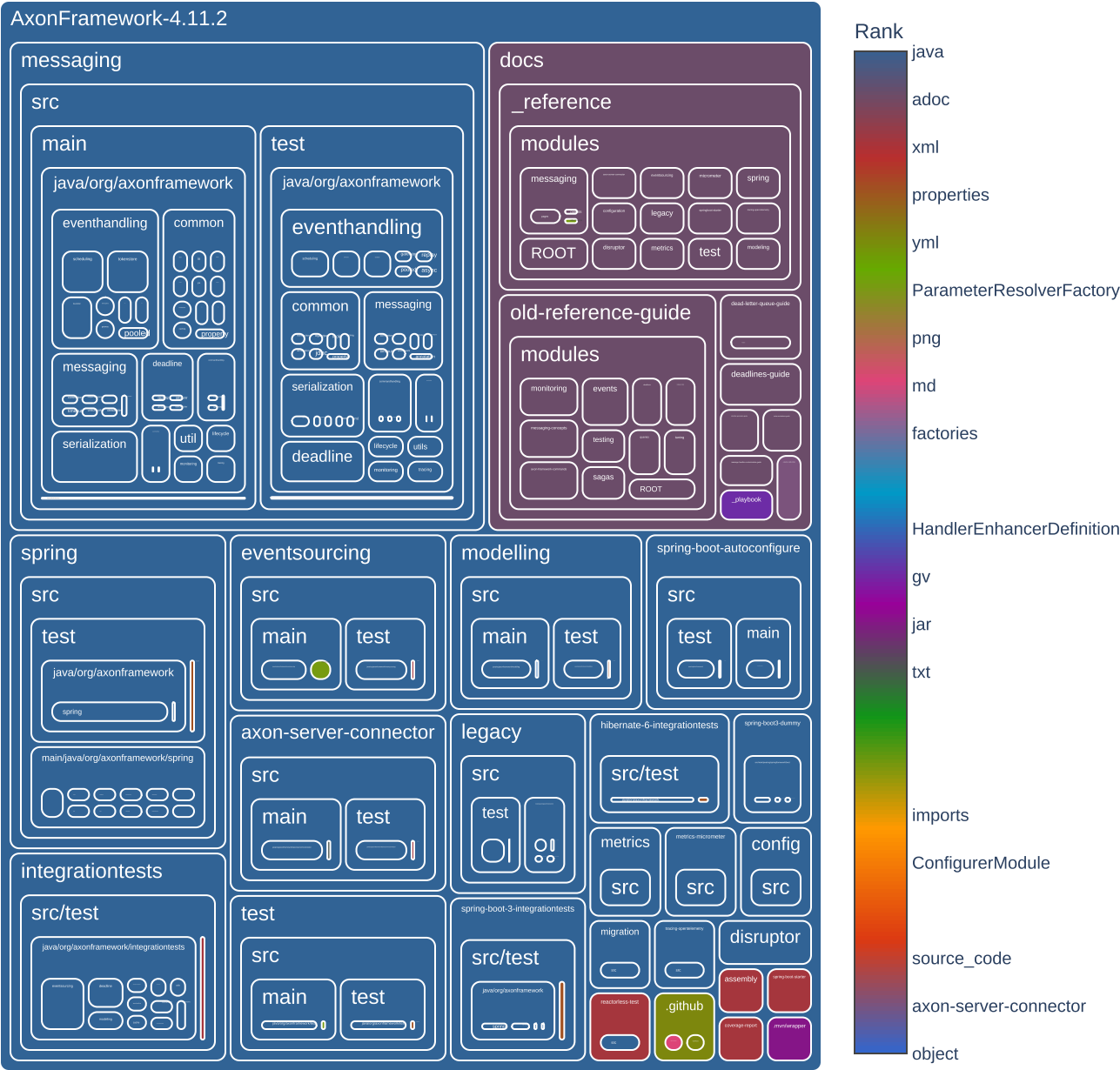
## Number of files per directory

## Directories and their file count



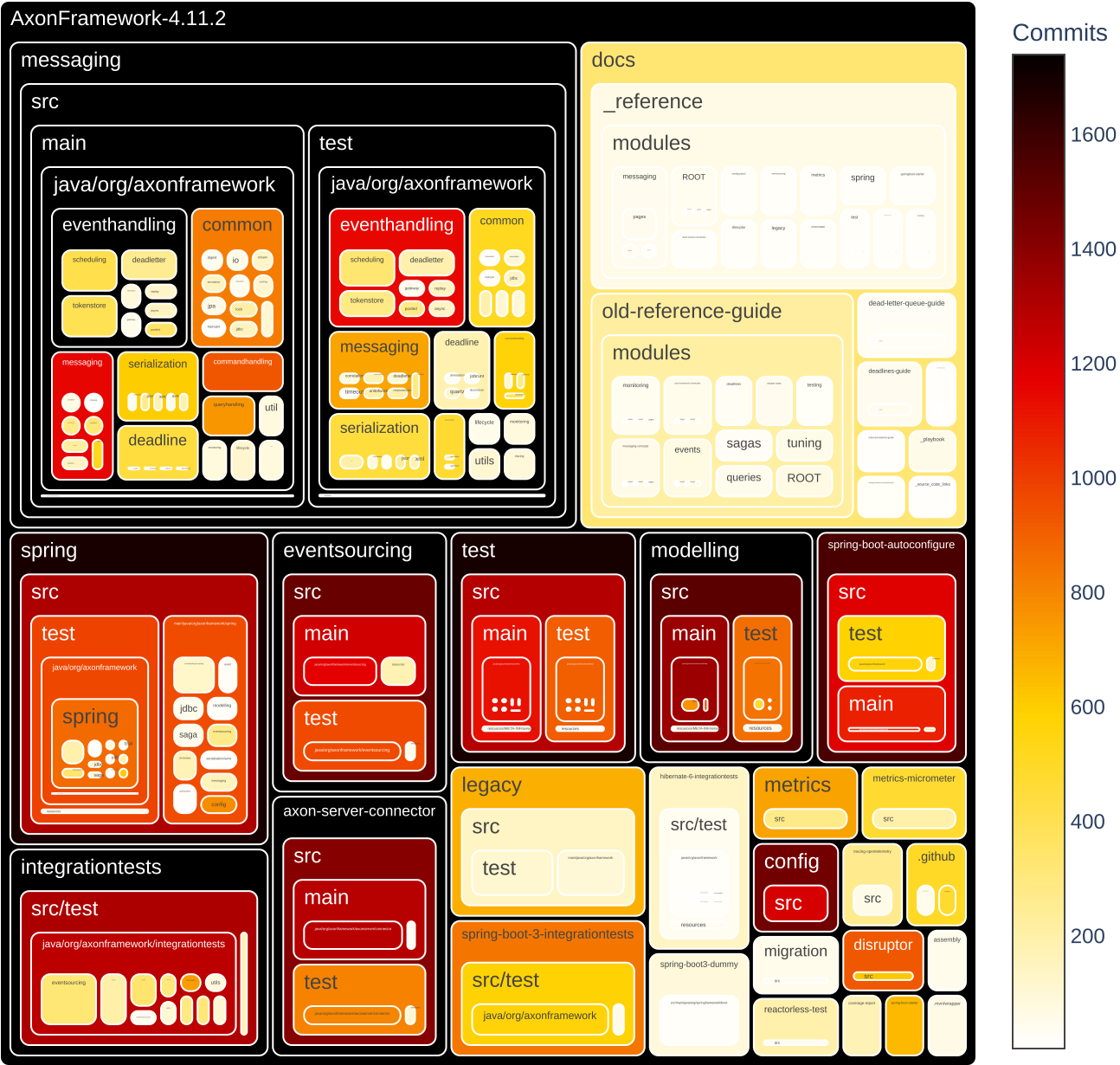
## Most frequent file extension per directory

Most frequent file extension per directory



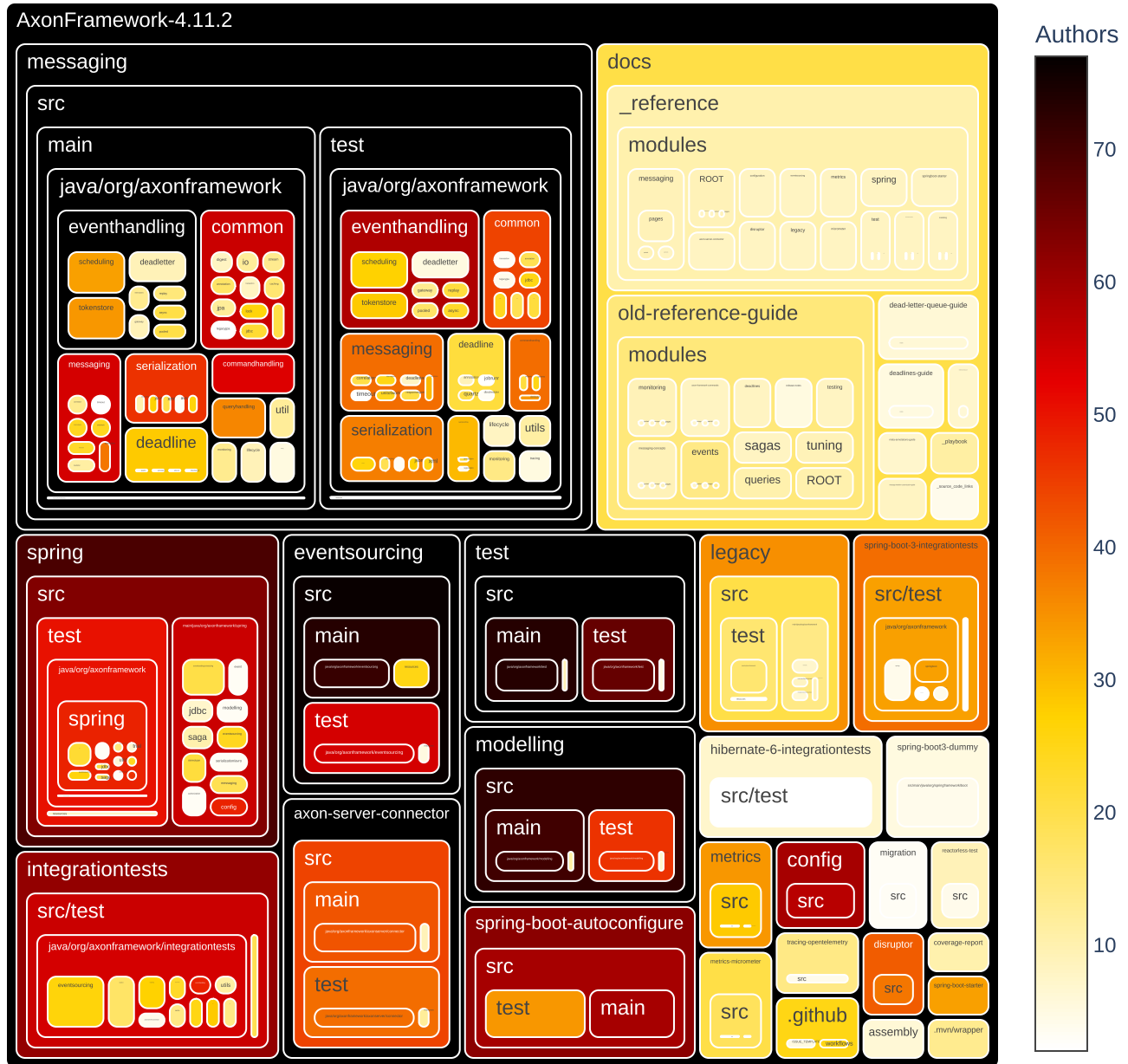
Number of commits per directory

Number of git commits



Number of distinct authors per directory

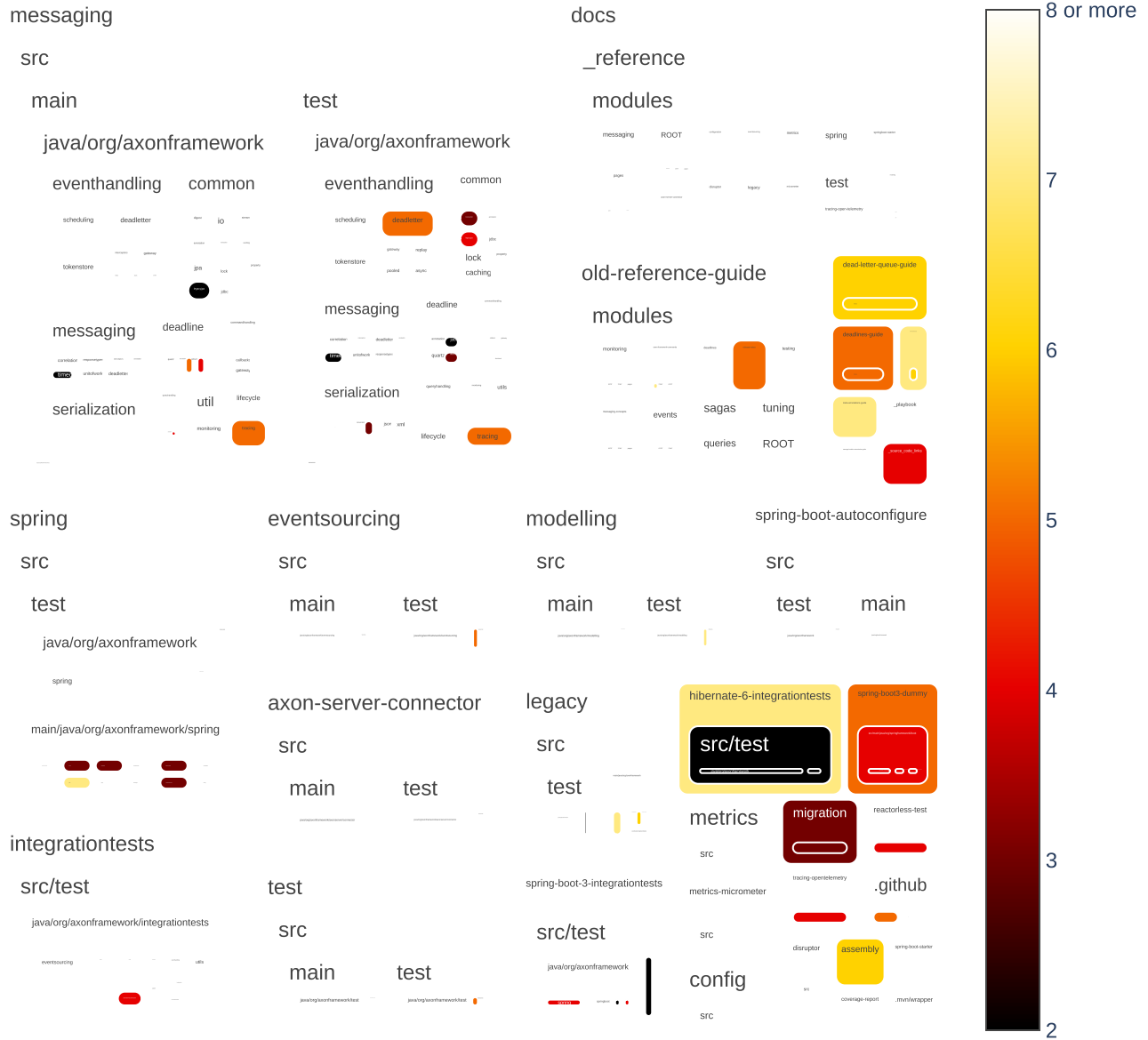
## Number of distinct commit authors



Directories with very few different authors

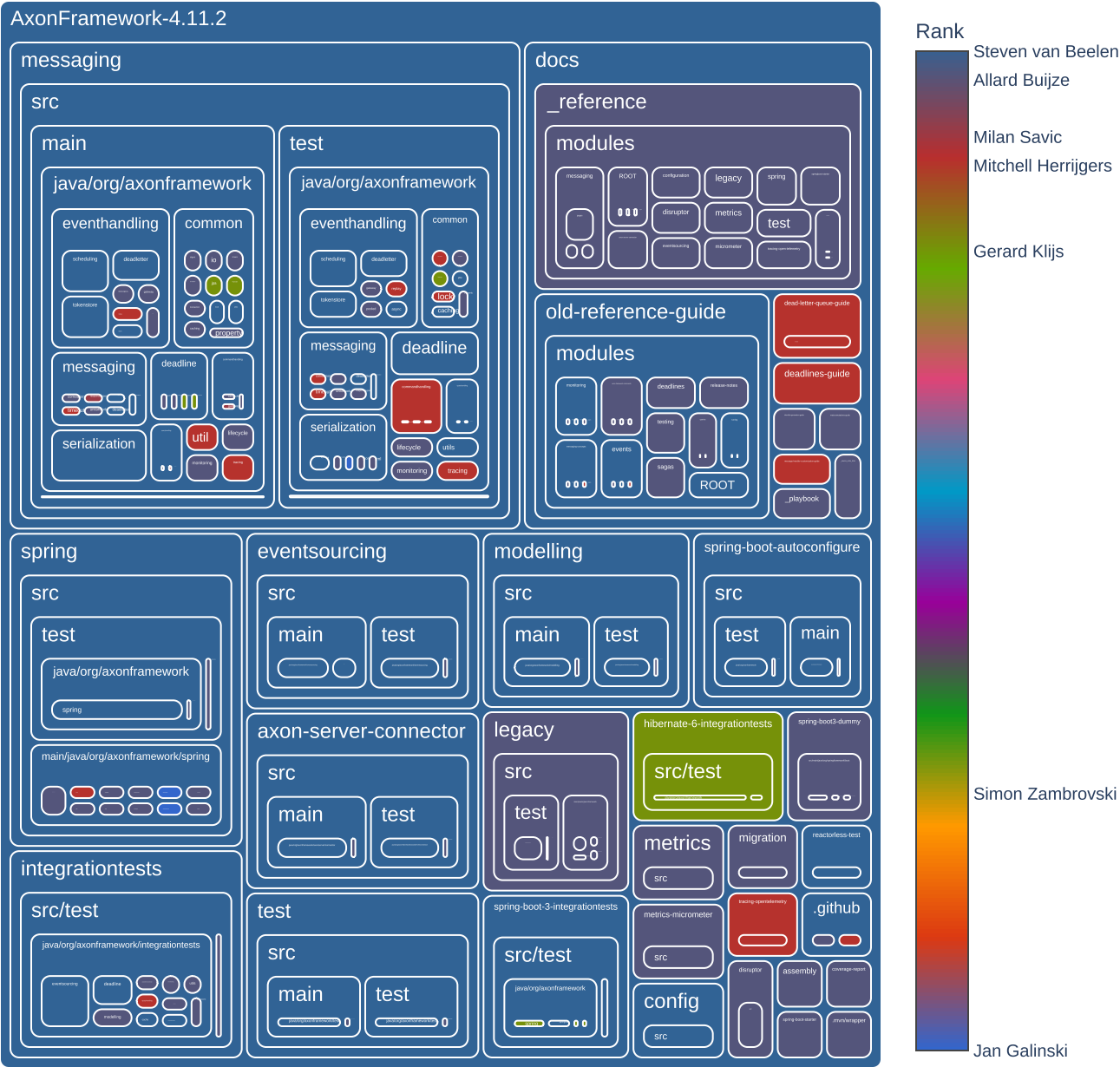
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.11.2



Main author per directory

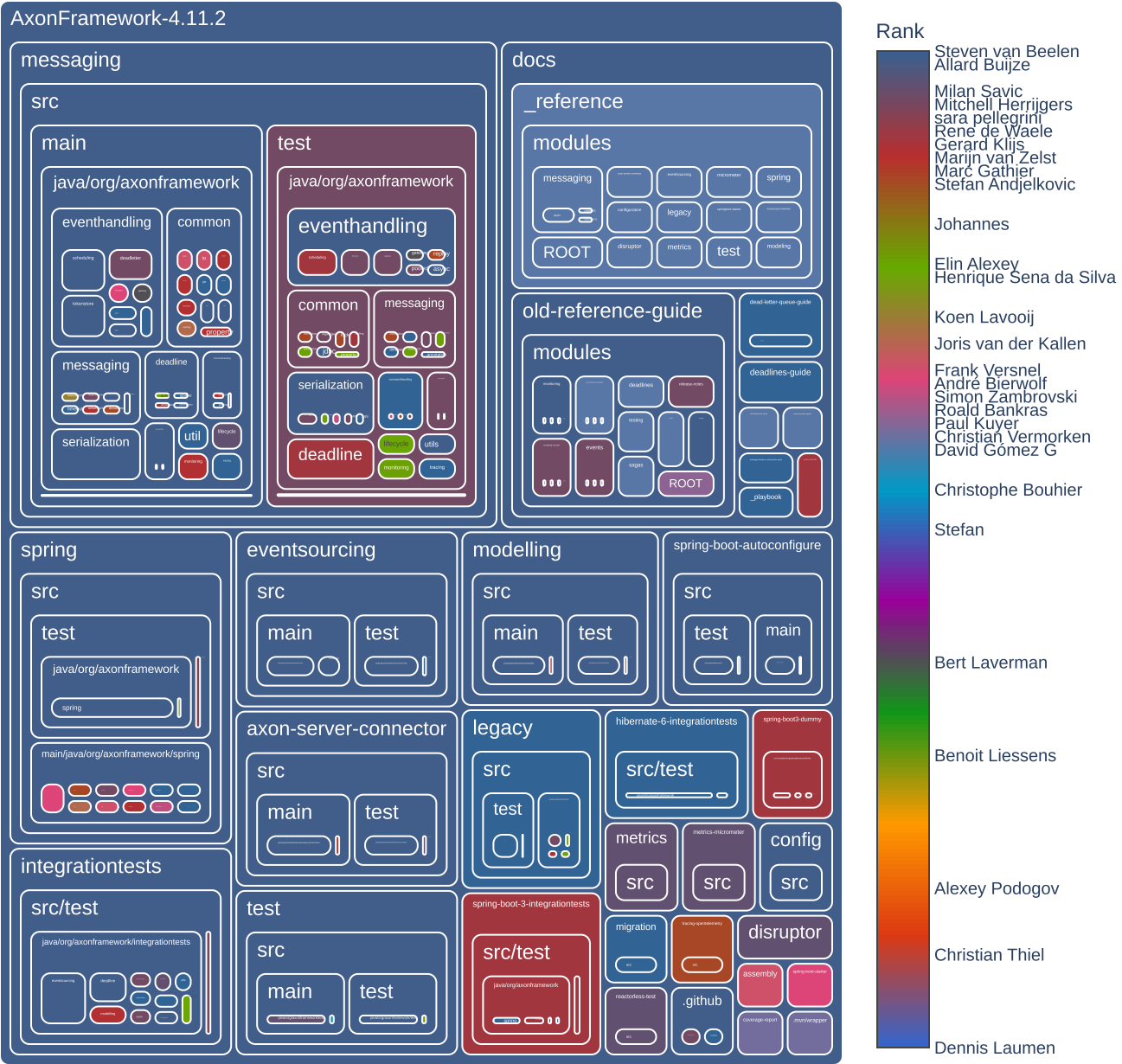
Main authors with highest number of commits



Second author per directory



Second author with the second highest number of commits

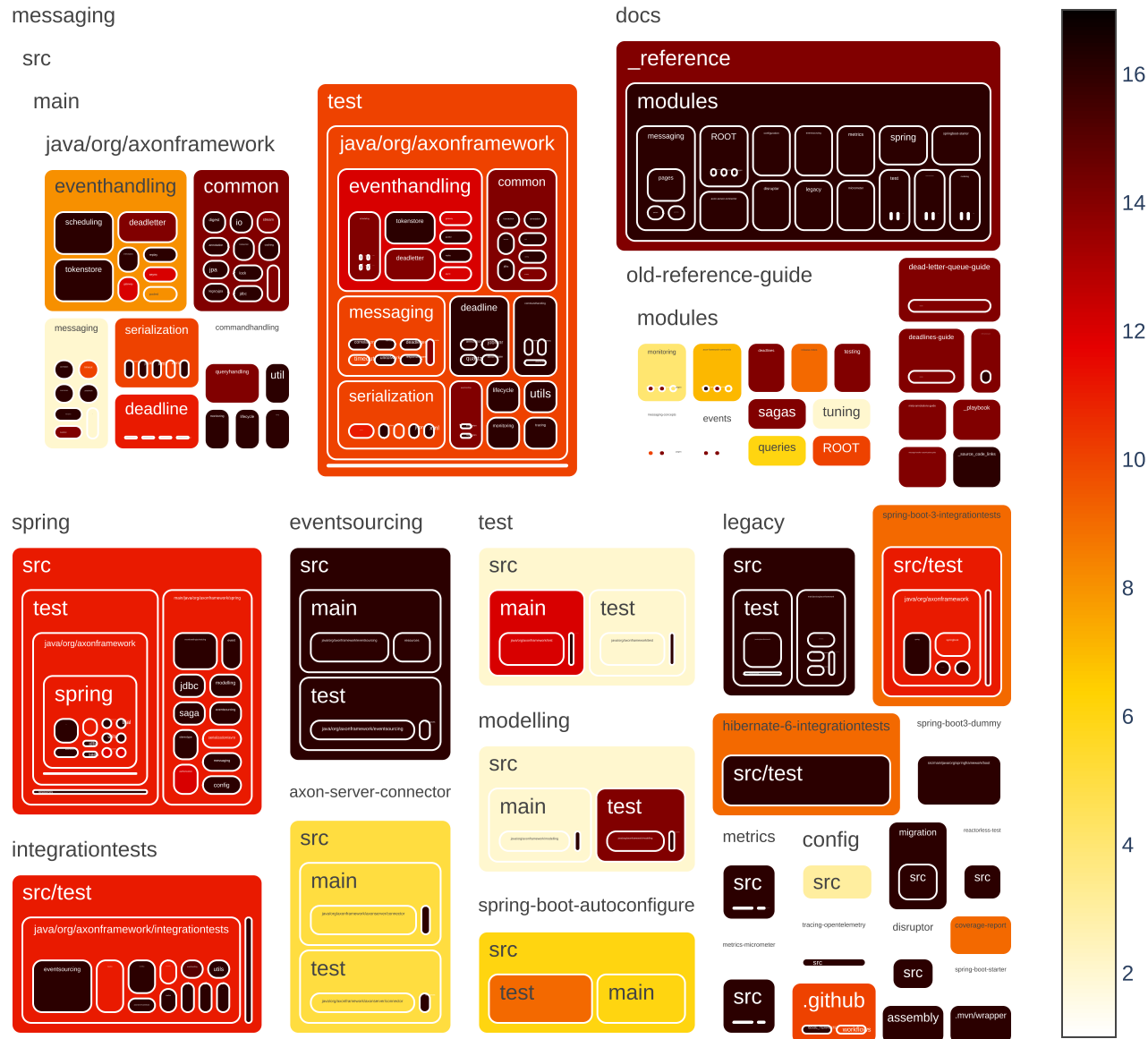


Days since last commit per directory



Rank of days since last commit

AxonFramework-4.11.2



Days since last file creation per directory

Days since last file creation

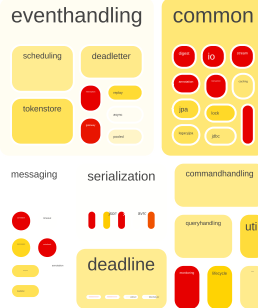
AxonFramework-4.11.2

messaging

src

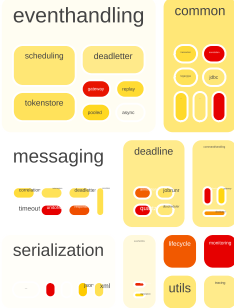
main

java/org/axonframework

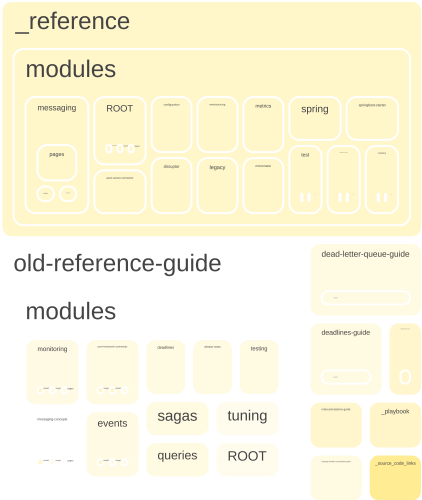


test

java/org/axonframework



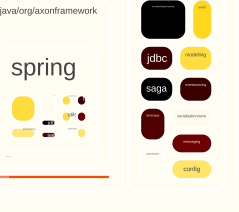
docs



spring

src

test



integrationtests

src/test



eventsourcing

src

main



src

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spring-boot-autoconfigure

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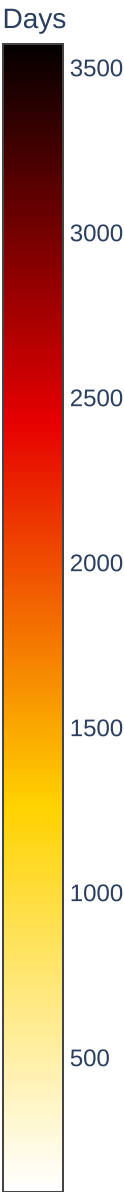
test

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test



Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-4.11.2



Days since last file modification per directory

Days since last file modification

AxonFramework-4.11.2

messaging

src

main

java/org/axonframework



test

java/org/axonframework



docs

\_reference

modules



old-reference-guide

modules



Days

2000

1500

1000

500

spring

src

test



eventsourcing

src

main

test



test

src

main

test



modelling

src

main

test



spring-boot-autoconfigure

src

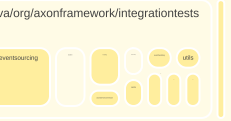
test

main



integrationtests

src/test



axon-server-connector

src

main

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legacy

src

test



hibernate-6-integrationtests

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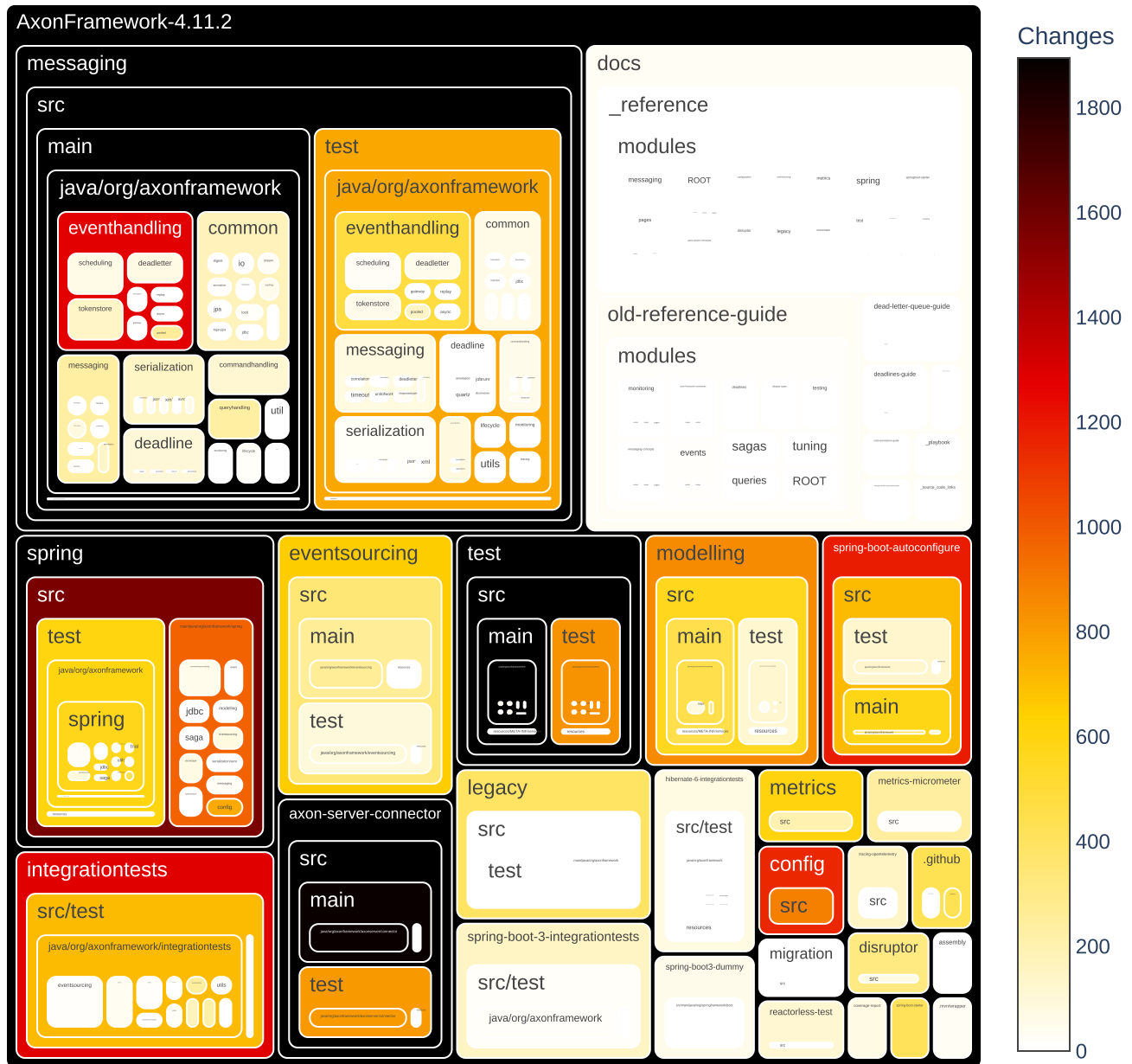
src

src

Days since last file modification per directory (ranked)



## Pairwise file changes



# Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

### Preview data

```
Sum of commits that changed more than 30 files (each) = 961
Max changed files with one commit = 2178
```

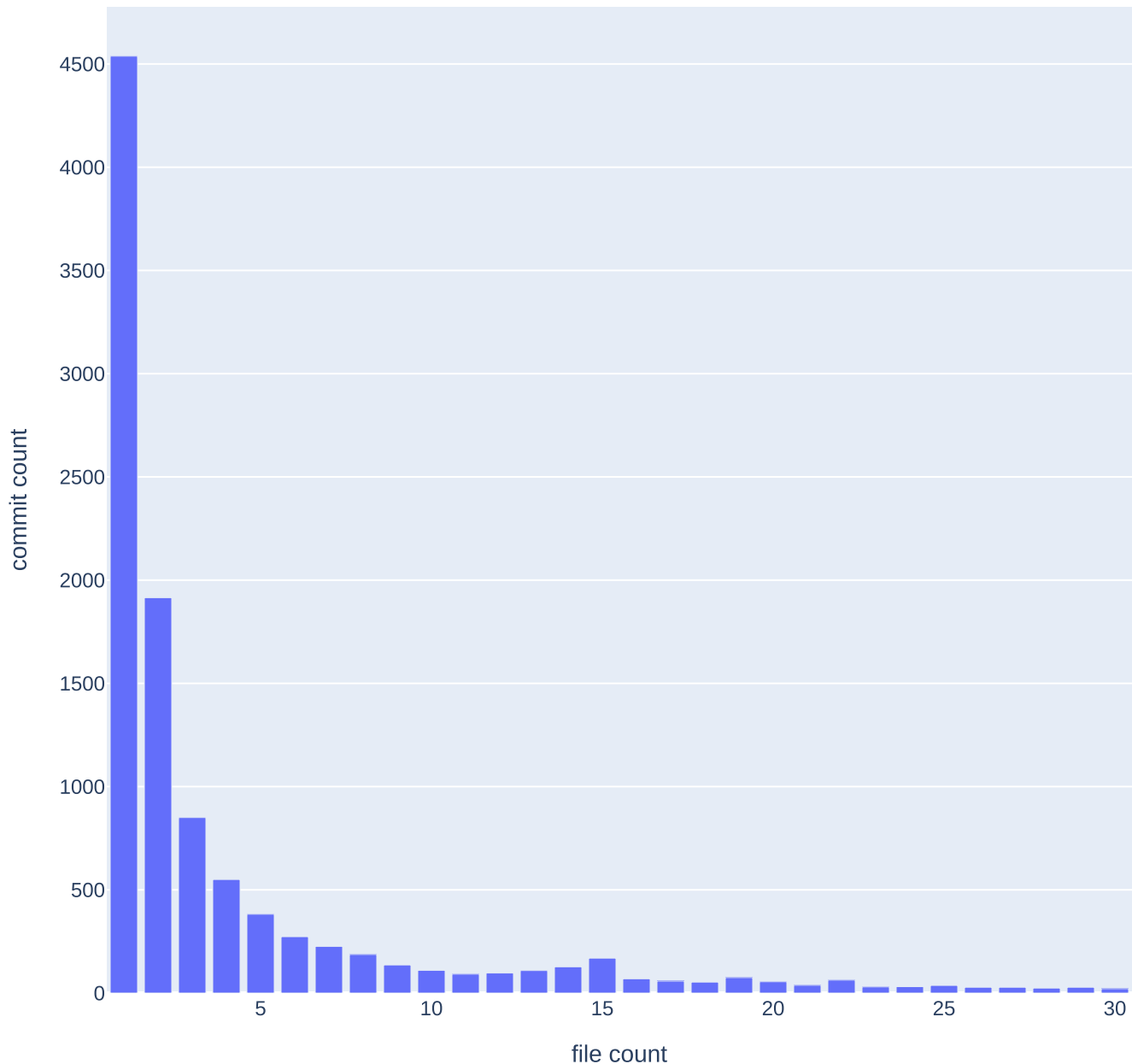


|       | filesPerCommit | commitCount |
|-------|----------------|-------------|
| count | 333.000000     | 333.000000  |
| mean  | 344.075075     | 34.114114   |
| std   | 440.375494     | 276.321358  |
| min   | 1.000000       | 1.000000    |
| 25%   | 84.000000      | 1.000000    |
| 50%   | 186.000000     | 2.000000    |
| 75%   | 428.000000     | 5.000000    |
| max   | 2178.000000    | 4538.000000 |

|    | filesPerCommit | commitCount |
|----|----------------|-------------|
| 0  | 1              | 4538        |
| 1  | 2              | 1915        |
| 2  | 3              | 850         |
| 3  | 4              | 550         |
| 4  | 5              | 382         |
| 5  | 6              | 273         |
| 6  | 7              | 226         |
| 7  | 8              | 187         |
| 8  | 9              | 136         |
| 9  | 10             | 110         |
| 10 | 11             | 92          |
| 11 | 12             | 97          |
| 12 | 13             | 109         |
| 13 | 14             | 127         |
| 14 | 15             | 169         |
| 15 | 16             | 69          |
| 16 | 17             | 58          |
| 17 | 18             | 53          |
| 18 | 19             | 75          |
| 19 | 20             | 55          |
| 20 | 21             | 38          |
| 21 | 22             | 63          |
| 22 | 23             | 30          |
| 23 | 24             | 31          |
| 24 | 25             | 37          |
| 25 | 26             | 28          |
| 26 | 27             | 28          |
| 27 | 28             | 24          |
| 28 | 29             | 28          |
| 29 | 30             | 21          |

Bar chart with the number of files per commit distribution

Changed files per commit



## Pairwise Changed Files vs. Dependency Weight

This section explores the correlation between how often pairs of files are changed together (common commit count) and their dependency weight. Note that these results should be interpreted cautiously, as comparing pairwise changes and dependencies is inherently challenging.

### Considerations

- **Historical vs. Current State:** Pairwise changes reflect the entire git history, while dependency weight represents the current state of the codebase.

- **Commit Granularity:** Developers may use different commit strategies, such as squashing changes into a single commit or creating fine-grained commits. Ideally, each commit should represent a single semantic change for accurate analysis.
- **Dependency Representation:** Some file types (e.g., Java files with import statements) clearly define dependencies, while others (e.g., shell scripts, XML, YAML) lack explicit dependency relationships.
- **Repository Characteristics:** Repositories with generated code may have many large commits, while stabilized repositories may only update configuration files for dependency changes.

## Data Preview

```
Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: cardinality)} {position: line: 9, column: 48, offset: 576} for query:
"// List pair of files that were changed together and that have a declared dependency between each other.\n\n MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n //De-duplicating the pairs of files isn't necessary, because the dependency relation is directed.\n //WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n WITH firstCodeFile.fileName AS firstFileName\n ,secondCodeFile.fileName AS secondFileName\n ,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n ,pairwiseChange.commitCount AS commitCount\n ,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\n RETURN dependencyWeight\n ,commitCount\n ,fileDistanceAsFewestChangeDirectoryCommands\n // ,count(*) AS occurrences\n // ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\n ORDER BY dependencyWeight, commitCount\n\n // MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n // MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n // WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n // RETURN firstCodeFile.fileName AS firstFileName\n // ,secondCodeFile.fileName AS secondFileName\n // ,dependency.weight AS dependencyWeight\n // ,pairwiseChange.commitCount AS commitCount\n // ORDER BY dependencyWeight, commitCount\n\n // MATCH (g1:!Git&File)-[relation:CHANGED_TOGETHER_WITH|DEPENDS_ON]-(g2:!Git&File) \n // WITH count(DISTINCT relation) AS relatedFilesCount\n // ,collect(DISTINCT relation) AS relations\n // UNWIND relations AS relation\n // WITH relatedFilesCount\n // ,coalesce(relation.commitCount, 0) AS commitCount\n // ,coalesce(relation.weight, 0) AS dependencyWeight\n // ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n // RETURN dependencyWeight\n // ,commitCount\n // ,fileDistanceAsFewestChangeDirectoryCommands\n // ORDER BY dependencyWeight, commitCount"
```

|    | dependencyWeight | commitCount | fileDistanceAsFewestChangeDirectoryCommands |
|----|------------------|-------------|---------------------------------------------|
| 0  | 1                | 12          | 0                                           |
| 1  | 1                | 12          | 0                                           |
| 2  | 1                | 12          | 0                                           |
| 3  | 1                | 12          | 2                                           |
| 4  | 1                | 12          | 0                                           |
| 5  | 1                | 12          | 4                                           |
| 6  | 1                | 12          | 4                                           |
| 7  | 1                | 12          | 0                                           |
| 8  | 1                | 12          | 0                                           |
| 9  | 1                | 12          | 4                                           |
| 10 | 1                | 13          | 0                                           |
| 11 | 1                | 13          | 4                                           |
| 12 | 1                | 13          | 0                                           |
| 13 | 1                | 13          | 4                                           |
| 14 | 1                | 14          | 2                                           |
| 15 | 1                | 14          | 4                                           |
| 16 | 1                | 15          | 2                                           |
| 17 | 1                | 15          | 0                                           |
| 18 | 1                | 15          | 0                                           |
| 19 | 1                | 16          | 0                                           |

## Data Statistics

'Pairwise changed git files compared to dependency weights - Overall statistics'

|       | dependencyWeight | commitCount | fileDistanceAsFewestChangeDirectoryCommands |
|-------|------------------|-------------|---------------------------------------------|
| count | 166.000000       | 166.000000  | 166.000000                                  |
| mean  | 9.566265         | 25.253012   | 1.373494                                    |
| std   | 19.257863        | 16.677579   | 1.570260                                    |
| min   | 1.000000         | 12.000000   | 0.000000                                    |
| 25%   | 1.250000         | 14.250000   | 0.000000                                    |
| 50%   | 3.000000         | 19.000000   | 0.000000                                    |
| 75%   | 10.000000        | 28.000000   | 2.000000                                    |
| max   | 154.000000       | 95.000000   | 4.000000                                    |

'Pairwise changed git files compared to dependency weights - Pearson Correlation'

|                                             | dependencyWeight | commitCount | fileDistanceAsFewestChangeDirectoryCommands |
|---------------------------------------------|------------------|-------------|---------------------------------------------|
| dependencyWeight                            | 1.000000         | 0.204462    | -0.143721                                   |
| commitCount                                 | 0.204462         | 1.000000    | -0.143874                                   |
| fileDistanceAsFewestChangeDirectoryCommands | -0.143721        | -0.143874   | 1.000000                                    |

'Pairwise changed git files compared to dependency weights - Spearman Correlation'

|                                             | dependencyWeight | commitCount | fileDistanceAsFewestChangeDirectoryCommands |
|---------------------------------------------|------------------|-------------|---------------------------------------------|
| dependencyWeight                            | 1.000000         | 0.212543    | -0.089280                                   |
| commitCount                                 | 0.212543         | 1.000000    | -0.132659                                   |
| fileDistanceAsFewestChangeDirectoryCommands | -0.089280        | -0.132659   | 1.000000                                    |

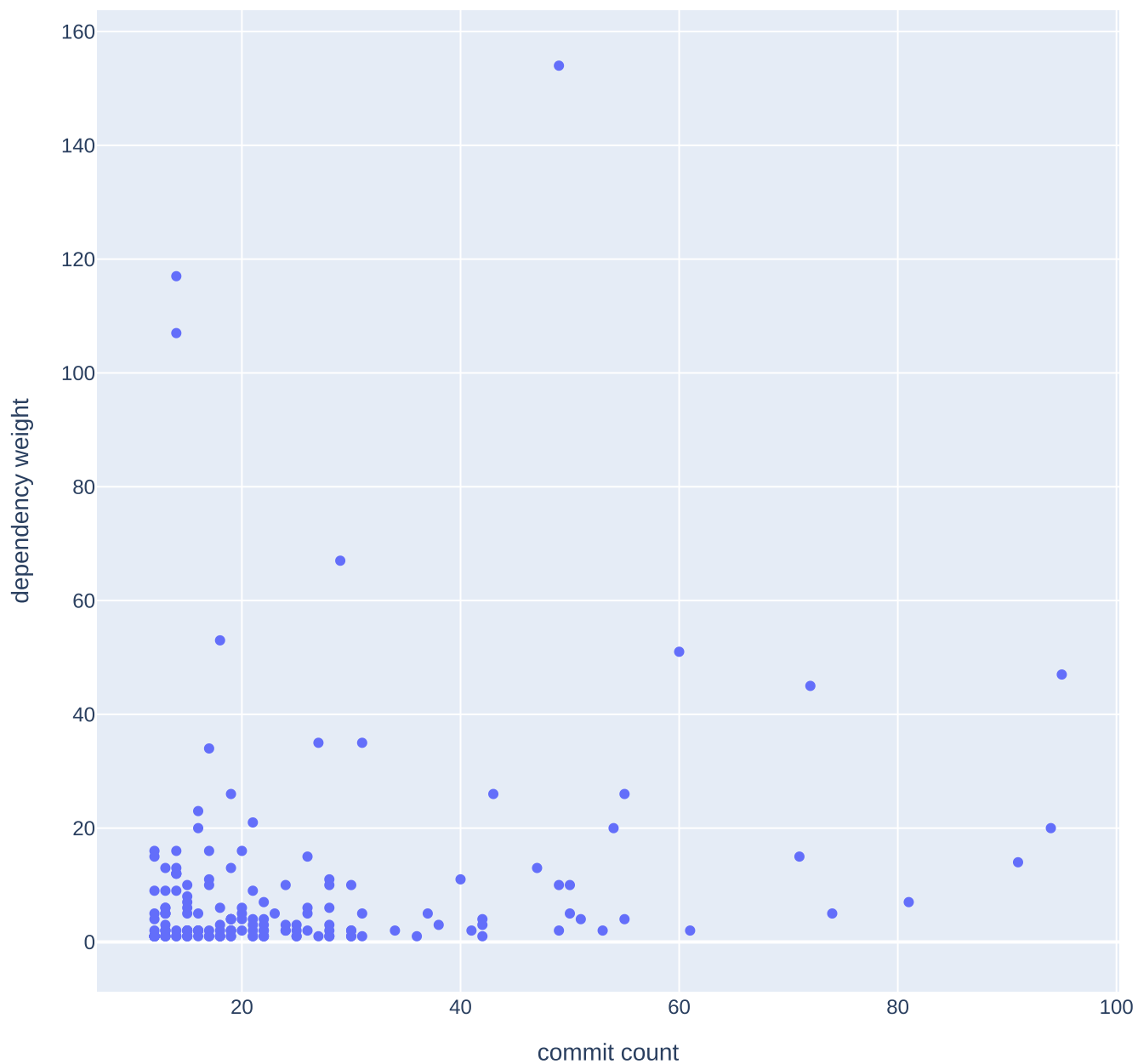
'Pearson Correlation with p-value for commitCount and dependencyWeight'

PearsonRResult(statistic=0.20446197739019414, pvalue=0.008232632527754543)

'Spearman Correlation with p-value for commitCount and dependencyWeight'

SignificanceResult(statistic=0.21254323842799402, pvalue=0.005973735857352672)

Pairwise changed files: Number of changes (commitCount) vs. dependency weight



WordCloud of git authors

|   | word                | frequency |
|---|---------------------|-----------|
| 0 | Steven van Beelen   | 3402      |
| 1 | Allard Buijze       | 2722      |
| 2 | smcvb               | 822       |
| 3 | Milan Savic         | 449       |
| 4 | Mitchell Herrijgers | 313       |
| 5 | Gerard Klijs        | 246       |
| 6 | Marc Gathier        | 231       |
| 7 | sara pellegrini     | 163       |
| 8 | Rene de Waele       | 121       |
| 9 | Jетро Coenradie     | 117       |

